

Request for Information (RFI)
Antibacterial Nanoparticles Therapeutic (ANT) Study

General Information:

This RFI is issued solely for information and planning purposes and does not constitute a solicitation. All information received in response to this RFI will be treated as business confidential information. Nonetheless, submitters should properly mark their responses if the information is confidential. Responses to the RFI will not be returned. Responses to this notice are not offers and cannot be accepted by the Government to form a binding contract. Responders are solely responsible for all expense associated with responding to this RFI.

RFI Response and due date: 1 April 2026

Requiring Activity Name:

Walter Reed Army Institute of Research (WRAIR), Bacterial Diseases Branch (BDB)

Contracting Office:

Defense Health Agency Contracting Activity (DHACA)

Description of Requirement:

The BDB seeks to conduct a study to test innovative antibacterial nanoparticles with phage-mimicking properties (PhANPs) against combat relevant multi-drug resistant (MDR) strains of *Pseudomonas* (P.) *aeruginosa*, bacterial pathogen to cause hard to treat and even deadly infections in service members after battlefield injuries. There is a need to bring novel therapeutics to the military health care to treat MDR *P. aeruginosa*. The collaborative partner of this study is the University of Notre Dame, the patent owner of a unique technology (phage-mimicking nanoparticles) that provides life-saving therapeutics for service members and civilians. This technology was awarded a patent in 2024 - US-12161725-B2. Government anticipates that the collaborative efforts will contribute to Food and Drug Administration (FDA) approval of this innovative therapy, leading to reduced morbidity and mortality rates, enhanced operational readiness, advanced combat medicine, and substantial contributions to broader biosecurity initiatives and civilian health improvements.

The contractor shall provide non-personal Research and Development effort. This includes qualified research personnel, appropriate laboratory facilities, and all necessary laboratory and research related materials and supplies.

North American Industry Classification System (NAICS) Code: 541713 - Research and Development in Nanotechnology

Specific Tasks and Details

General: The contractor shall complete three milestones, as defined below, as part of an antibacterial nanoparticles therapeutic study with WRAIR, BDB. The contractor shall provide all qualified research personnel, facilities, laboratory equipment, and essential materials and supplies to complete study milestones.

Study Background: In recent military conflicts, skin and soft tissue infections were predominant but life-threatening systemic infections and sepsis were also detected frequently. In combat casualties, which are often characterized by blast and burn injuries, subsequent infections with *Klebsiella pneumoniae* or *Pseudomonas aeruginosa* are frequently associated with mortality. Additionally, most pathogens isolated from military personnel are MDR, and the need for new treatments to target MDR pathogens is enormous. This study proposes to test innovative PhANPs against combat-relevant strains of *P. aeruginosa*. Aim One will assess if diverse combat-relevant MDR *P. aeruginosa* strains are sensitive to the nanoparticles. Aim Two will test the efficacy of PhANPs against the most prevalent *P. aeruginosa* strains in a simple pre-clinical model of *Galleria mellonella* (wax-worm). In Aim Three, the novel treatment will be evaluated against lethal and chronic wound models of *P. aeruginosa* infection in mice. Additionally, PhANPs efficacy will be assessed against a sustained wound *P. aeruginosa* infection. In Aim Four, the novel treatment will be evaluated in wound swine model. In all animal models of infection, the study will evaluate synergy of nanoparticles with antibiotic ceftazidime. The studies aim to bridge the gap between laboratory discoveries using pre-clinical small and large animal wound infection models and practical applications in the field. It will evaluate preclinical efficacy of treatment drug in validated animal model against MDR pathogens. Ultimately, this innovative therapy could lead to reduced morbidity and mortality rates, enhanced operational readiness, and advanced combat medicine.

Contractor Tasks: This study shall be broken down into **three specific milestones**. Tasks are specified below for each milestone. The contractor shall have capability to produce multiple batches of nanoparticles with identical and verifiable characteristics to ensure the long-term reproducibility of the research. Subsequent milestones are not to be initiated until the previous milestone is complete and/or approved by the COR to proceed to the next milestone tasks.

Milestone Tasks

Milestone One Tasks: SilverGold Nanoparticles shall be produced and conjugated to antimicrobial peptide/polymer for testing against *P. aeruginosa* biofilms produced in wound-like medium.

- Quantity: 17 batches estimated = 500 mL of 200 μ M stock PhaNP@Syn20.
- Deliverable: The 500 mL of 200 μ M stock PhaNP@Syn20 shall be aliquoted, lyophilized, and shipped on ice. Aliquots shall be resuspended to the desired stock concentrations for further testing. Properties of nanoparticles for each batch shall be assessed using precision methods of physical chemistry. Biological properties of each batch shall be determined internally against *P.aeruginosa FRD1* and Methicillin-resistant *Staphylococcus aureus* (MRSA) strains for quality control (QC) benchmarking before shipping to WRAIR. QC certificate shall be provided to WRAIR using Technical Exhibit
- Timeline: Months 0-9

Milestone Two Tasks: SilverGold Nanoparticles shall be produced and conjugated to antimicrobial peptide/polymer for testing against *P. aeruginosa* wax-worm infection.

- Quantity: 3 batches estimated = 75 mL of 200 μ M stock PhaNP@Syn20.
- Deliverable: 75 mL of the 200 μ M stock of PhaNP@Syn20 shall be aliquoted, lyophilized, and shipped on ice. Aliquots shall be resuspended to the desired stock concentrations for further testing. Properties of nanoparticles for each batch shall be assessed using precision methods of physical chemistry. Biological properties of each batch shall be determined internally against *P.aeruginosa FRD1* and MRSA strains for QC benchmarking before shipping to WRAIR. QC certificate shall be provided to WRAIR using Technical Exhibit
- Timeline: Months 10-18

Milestone Three Tasks: SilverGold Nanoparticles shall be produced and conjugated to antimicrobial peptide/polymer for preparation of batch for all animal work.

- Quantity: 20 batches estimated = 570 mL of 200 μ M stock PhaNP@Syn20.
- Deliverable: The 570 mL of 200 μ M stock PhaNP@Syn20 shall be aliquoted, lyophilized, and shipped on ice. Aliquots shall be resuspended to the desired stock concentrations for further testing. Properties of nanoparticles for each batch shall be assessed using precision methods of physical chemistry. Biological properties of each batch shall be determined internally against *P.aeruginosa FRD1* and MRSA strains for QC benchmarking before shipping to WRAIR. QC certificate shall be provided to WRAIR using Technical Exhibit
- Timeline: Months 19-36

QC Certificates. The contractor shall provide documentation evaluating the following fundamental properties as well as the effectiveness of nanoparticles in a controlled laboratory setting against the specified reference strains.

Fundamental Properties: The contractor shall provide comprehensive documentation demonstrating the material's fundamental properties. This shall include verifying the nanoparticle's chemical purity and composition, while also identifying any residual reagents from the synthesis process. A detailed analysis of the average particle size, size distribution, and morphology is required, supported by techniques such as Transmission or Scanning Electron Microscopy. Furthermore, data shall be provided on the nanoparticle's stability and suspension characteristics in relevant solutions, which is crucial for ensuring accurate and repeatable experimental results.

Effectiveness: The contractor shall provide documentation evaluating the nanoparticle's direct antibacterial effectiveness in a controlled laboratory setting against the specified reference strains. This *in vitro* assessment shall establish a baseline for the nanoparticle's antimicrobial efficacy. Key metrics for this evaluation shall include determining the Minimum Inhibitory Concentration (MIC), which is the lowest concentration that prevents visible bacterial growth, and the Minimum Bactericidal Concentration (MBC), the lowest concentration that results in bacterial death. The evaluation shall also measure the time-kill kinetics to understand how quickly the nanoparticles eliminate the bacteria and shall test the spectrum of activity against reference Gram-positive and Gram-negative bacteria to determine the breadth of the nanoparticle's effectiveness.

Shipping of Deliverables: All electronic documents shall be delivered to the COR as specified on the deliverables table. Physical deliverables shall be packaged as specified and shipped to the address listed below. Costs associated with packaging and shipping are the responsibility of the contracted vendor, as included in the contract cost.

Place of Performance: Contractor Facility

Instructions to Responders:

Responses from all capable and qualified sources are encouraged to respond to this request.

White papers should adhere to the following formatting and outline instructions:

1. All submissions shall be made electronically and be in one of the following formats: Microsoft Word or Adobe PDF.

2. Cover Page (1page)

- Title
- Organization
- Responders technical and administrative points of contact (names, addresses, phone, fax number and email addresses)
- Topic area(s) addressed
- NAICS Code

3. Interested parties are asked to respond to this RFI and submit a capabilities statement that supports the focus of this RFI as detailed below. Please review the place of performance requirements. Please submit written responses via email in Microsoft Office 2000 or 2007 format or Adobe PDF. All responses should be complete: in 12-point font; and not to exceed 3 pages total, including graphics, tables, and appendices.

No hardcopies shall be accepted. No written solicitation document is available at this time. Telephone requests will not be honored, and no bidders list shall be maintained. No inquiries regarding any potential future acquisition activities shall be entertained.

Potential offerors are requested to direct all questions via email to the Point of Contact listed below. Responses are due to the Point of Contact, Ms. Linda Hinson at email: linda.t.hinson.civ@health.mil and Mr. Sharew Haily at email: sharew.hailu.civ@health.mil no later than 10:00AM ET, 1 April 2026.